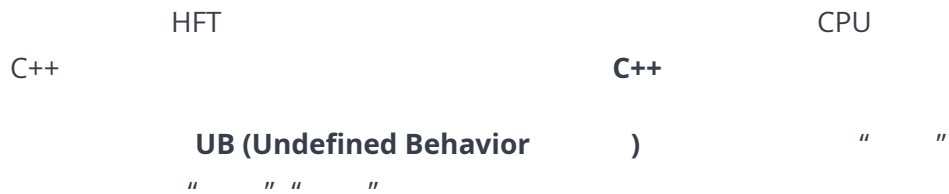


C++ UB

" "

A Brief Talk on C++ UB: Math, Hardware, and Compiler Lies



1.

```
bool check_overflow(int x) {  
    if (x + 1 < x) {  
        return true; // / Overflow wrap-around occurred  
    }  
    return false;  
}
```

" " `x + 1 < x` 32

GCC Clang -O2 -O3

```
check_overflow(int):  
    xor     eax, eax  
    ret     ; false / Returns false directly!
```

2. " " vs

C++ " "

(Unsigned)

`unsigned int` C++ Modulo Arithmetic
`x + 1 < x` " "

(Signed)

C++ `int` CPU
UB
UB Optimizer
1. UB
2.
3.
4. `x + 1 < x` Dead Code

3. C++20 " "

CPU x86, ARM ** Two's
Complement **
C++20
C20 ** C **
UB
UB
GCC LLVM

- " "
- " `for (int i = 0; i <= n; ++i)` Loop
- Unrolling SIMD `i` "
- " UB C++ "

4. C++

C++ " "

CPU

C++

" "